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FROM CLICKSTREAM DATA TO PURCHASE INTENTION

The Role, Business Impact and Challenges of Predictive Analytics in E-Commerce

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1. Introduction

E-commerce platforms continuously record a stream of interactions as visitors move between pages, visualize products, compare alternatives and either complete/abandon a purchase transaction. These records create good opportunities to observe online users behaviour at a level of detail that conventional online surveys like Customer Satisfaction Survey (CSAT), Net Promoter Score Survey (NPS), Checkout Experience Survey, Cart-Abandonment Survey and aggregated sales reports generally cannot provide. However, the records alone do not generate managerial insights. Big Data Analytics (BDA) can be better understood as a combination of data, analytical methods, technologies, people and organisational processes through which evidence is converted into business and strategic decisions and, not only a synonym for huge datasets or sophisticated software (Akter & Wamba, 2016). Data Science powerfully supports this process by combining various elements like statistical reasoning, computational methods and domain knowledge to identify behaviour and patterns, build predictive models and evaluate whether their outputs are useful for e-comerces and businesses.

The business application of digital data may be analysed by reference to the attributes of volume, velocity, variety, veracity and value. E-Commerce environments may indeed generate massive volumes of rapidly changing transactional behavioural, textual and contextual data. However, the relevance of these dimensions varies upon the operational setup. A set of examples does not become strategically important merely by collecting many examples, neither ought an educationally motivated website session sample to be presented as a large scale and high velocity big data system. The more compelling contrast can be drawn between the restricted dataset allowing for demonstration of an analytical approach and the broad organisational set up in which the events may be continuously sampled over customers, devices and channels. In a recent overview both structured and unstructured traces contribute to consumer big data whereas data quality, interpretability and privacy remain constant impediments toward their effective utilisation (Liu et al., 2026).

Analytics can be descriptive, predictive or prescriptive. Descriptive analytics reflects on what has occurred during the customer journey. For example, it may produce a report on where customers came from, which pages and products they viewed, how they navigated the website and at what point they left or purchased products or services. Unlike descriptive analytics, predictive analytics can estimate a possible scenario or even an unobserved result from the available evidence. Prescriptive analytics, like predictive analytics, examines data but recommends an action, for example whether an incentive, product recommendation or service should be offered to the client. Although the three types have the capability to support each other in different circumstances, they are also used for different purposes.

For example, if a session receives a high purchase score, whether to offer a discount, a new service or simply do nothing should be left to the sales managers. That is why such actions must actually be tested and the results reviewed by the sales managers or other decision-makers. The score also does not imply that performing any of these actions would cause the customer to buy.

Clickstream data refers to the sequence or aggregation of a visitor's interactions with an e-commerce website or service. One can use categories of pages visited, visit length and navigation patterns, traffic source, device and session context to signify engagement. Purchase-intention prediction applies supervised classification to these signals to distinguish those sessions that are likely or unlikely to result in a purchase. One has to be careful with the terminology, because when the target is a completed purchase, the models do not predict the visitor's actual intention. Instead, they predict an observed session outcome based on its different features. For this reason, intent prediction, buying-session prediction and purchase-decision prediction have been identified as related but different prediction tasks (Cirqueira, Hofer, Nedbal, Helfert, & Bezbradica, 2020).

Additionally, time is important. If classification is performed after the session has ended, one can illustrate what behaviours were associated with conversion, but cannot intervene in that session. Making the prediction early is preferred, although fewer behavioural signals are available then. Requena, Cassani, Tagliabue, Greco and Lacasa (2020) explicitly separate full-session classification from early-prediction settings, showing that with limited clickstream information one can predict purchases. This introduces a fundamental trade-off: waiting until richer evidence is available may improve the prediction, while predicting earlier preserves the opportunity to influence the customer journey. The statistical importance of variables should be checked, as well as whether those variables would be available at the decision point; otherwise, these variables cannot contribute to increasing decision quality.

With an increase in focus around purchasing prediction, the organisational motivation has obvious merits. Behavioural estimates could aid an e-commerce company in prioritising assistance, tailoring recommendations, allocating promotional resources or reducing irrelevant intrusions. But is technical capability by itself enough? The evidence linking Big Data Analytics capability to firm performance shows that value also depends on the organisation's ability to reconfigure processes and act on analytical insight (Wamba et al., 2017). Most important, collecting and combining behavioural signals also poses significant privacy risks, increased data management responsibilities and concerns about loss of customer control. The questions of prediction effectiveness, relevance for the business and ethical acceptability cannot, thus, be considered on their own, separately.

How can clickstream data and predictive analytics support session-level purchase prediction and what business, methodological and ethical challenges affect their effective use in e-commerce? The goal is to critically examine how behavioural data can inform session-level purchase prediction and the conditions under which analytical outputs can generate business value. Prediction timing, class imbalance, choice of metric, interpretability are addressed. Privacy and governance, as well as model performance and organisational impact are considered

The motivation for the study is addressed after this introduction. Furthermore, the issues related to Big Data and Data Science in e-commerce, clickstream based insight, purchase prediction, business impact as well a list of possible methodological difficulties and ethical challenges and on turning analytical insight to value are discussed in the thematic review section. Subsequently, the discussion develops a decision-oriented framework and practical suggestions, and the paper concludes by providing answers to the research question and identifying the main implications.

2. Motivation

Purchase prediction explores an imbalance which is present in the area of online retail; many sessions result in browsing activity and few sessions result in revenue generation. In the Online Shoppers Purchasing Intention dataset used for the practical component of the Big Data and Data Science course, 1,908 out of the 12,330 sessions result in purchasing activity, which represents approximately 15.5% (Sakar & Kastro, 2018). This imbalance, which may not itself indicate a management problem, also implies a decision-making issue: identifying the minority of promising sessions without affecting the majority of non-purchasing visitors through discounts, messages or intrusive personalisation.

Because purchases are rare, most sessions may be classified by a model as non-purchasing rather than purchasing, leading to a seemingly satisfactory result in terms of accuracy. The minority class must be considered instead to evaluate the distribution of errors and their business consequences, because errors in classification may affect profit margins: false negatives may refer to lost opportunities to cater to customers who were likely to purchase, whereas false positives might refer to granting a discount or attention to a session which would not result in a purchase. The relative costs of errors vary for specific interventions and affect decisions concerning metric choice, thresholds and the way the intervention is designed.

Further, information richness in customer behaviour and the responsible use of data pose a constant tension. Richer information about past customer behaviour, additional outside attributes and cross-device identifiers may enable better profiling of customers, but they may also lead to customer privacy concerns and governance issues. Consequently, the business case for personalisation is conditional, as customer trust may erode even if an intervention is statistically well-targeted due to excessive data collection or messaging. Akter and Wamba (2016) identify privacy, security, informed consent, skills and governance as challenges for e-commerce analytics. These issues affect value creation rather than being treated only as compliance requirements.

The topic of Data Science within e-commerce forms an interesting bridge between discussions on Big Data strategy and controlled application studies based on real-world e-commerce datasets. The practical application uses an interpretable classification method - a decision tree - on a small well-documented public data set. It illustrates the reasoning process from a business question to data preparation, evaluation of the model and interpretation of its results. The practical study does not claim that 12,330 sessions represent Big Data at the scale of a contemporary e-commerce company. It also does not reproduce the infrastructure used by a large retail enterprise. However, it demonstrates the reasoning necessary to apply Data Science to marketing challenges through simple and understandable methods. Further on, the paper investigates the measures and infrastructure required in a large-scale e-commerce environment. This involves data collection, organisational capabilities, privacy safeguards and model monitoring. Evidence is also needed to determine if decisions supported by Data Science models are improving business outcomes.

3. Review of the Chosen Thematic

3.1 Big Data and Data Science in E-Commerce

Since business activities and customer interactions take place through digital systems which can record behaviour while it is unfolding, e-commerce may be a relevant setting for Big Data Analytics. These systems can accumulate transaction histories, website navigation, product searches, customer reviews, traffic sources and information about the context in which visits took place. Not surprisingly, analysis can go beyond reporting completed sales. Behavioural traces can help firms examine how customers reach a decision, the difficulties they face, which products attract attention but are rarely purchased, how they compare products, which actions tend to precede a purchase, and much more. Liu et al. (2026) provide a systematic review of 127 studies on the topic to show what kinds of data are being used in consumer Big Data research: both structured and unstructured. Liu et al. further show that consumption patterns, sentiment and decision pathways are being studied. Yet these records merely reflect traces of digital behaviour on the web rather than complete accounts of customer needs and motivations.

Volume, velocity, variety, veracity and value: this is typically the most frequently heard list of characteristics used to explain why Big Data is relevant for e-commerce. Volume concerns the potentially large number of customers, sessions and events. Velocity corresponds to the speed with which interactions are generated and may need to be analysed. Variety means that numerical, categorical, textual and sequential data are combined with each other. Veracity means that records need to be accurate, consistent and appropriate for the analysis that will be based on them. Finally, value is obtained from the resulting analysis when a decision or process is improved. These qualities are interdependent - more volume or richer variety give scope for additional insights, or the possibility to look more closely into the data than before, but can also intensify integration, quality and governance problems. Scale is not how Big Data should be interpreted, because it is one characteristic rather than the defining attribute. For additional clarity, BDA for e-commerce should be understood as a wider system involving data, technologies, analytical capabilities, people and organisational processes (Akter & Wamba, 2016).

These resources can be examined and turned into evidence using Data Science methods such as data preparation, exploratory data analysis, feature engineering, statistical modelling, machine learning and communication of results. There are examples of these applications in e-commerce, including understanding historical events through descriptive analysis, predicting outcomes such as purchases, abandonment and customer responses through predictive analysis, and subsequently designing possible courses of action through prescriptive analysis. A predictive model may predict a result, but it does not specify what should be done with the outcome. This leap has to be made with consideration of time, cost, effort, customer experience and organisational objectives. Even if a prediction is appropriate, it may arrive too late to be fed into an operational process, if one is available. It also does not explain what the intervention should be, and this decision needs to be taken by an organisational team using additional available information. Technically correct work may not be useful or valuable when it becomes available too late, when the prediction cannot be integrated into existing processes or systems, or when there is no clear business decision being supported.

The organisational capability to extract value from analytics is as important as technical capability. Based on survey results from 297 managers and business analysts, Wamba et al. (2017) identified relationships between Big Data Analytics capability, process-oriented dynamic capabilities and firm performance. The results suggest that organisations can benefit not just from having analytical resources but also from using insights to respond to changes and adapt processes. However, the study relied on cross-sectional and perceptual measures from a single country context; therefore, universal or causal inferences are limited. This evidence supports the importance of organisational capability but does not ensure that investments in analytics improve the performance of all e-commerce firms.

It is also necessary to distinguish between the operational scale of major platforms and the datasets commonly used in research or teaching. A large retailer might process continuous streams of website activity across millions of customers, whereas public datasets will often contain aggregated and temporally bounded observations. These smaller-scale datasets may still be useful to show Data Science methods, but should not be framed as similar to the infrastructures on which present-day Big Data is based. They are useful for clearly understanding how a business question can be translated into available or constructed variables, a model and an evaluation process. When these datasets are placed into the wider Big Data picture, they raise further questions about scalability, data integration, organisational readiness and governance. Clickstream data bridge online behaviour and purchase prediction, but their meaning and usefulness depend on how sessions, events and the intended prediction point are defined.

3.2 From Clickstream Data to Customer Insight

Clickstreams are records of the user's interaction with the online platform. Events may record page views, searches, category views, clicks, additions to or removals from the cart, and transactions. The time and sequence of each event are also registered. Related events are grouped into a session to represent a behavioural journey. This session can be represented as aggregated variables or as the original sequence of events. The kind of representation directly impacts on what the analysis can detect and how quickly an insight can be produced.

Clickstream data are valuable because they record observed behaviour rather than relying completely on recollected reports from customers. Navigation, engagement and hesitation patterns may indicate whether the visitor is progressing towards or away from a transaction. However, these patterns should be interpreted with some caution. Increased browsing time, repeat product views or cart activity can be related to interest, but they can also indicate confusion, comparison or difficulty with using the site. These behavioural records thus represent signals from which customer insight can be inferred; however, motivations are not revealed with certainty. As Liu et al. (2026) observe, digital traces can allow detailed analysis of consumer decision pathways, but data quality and the relationship between behaviour in the traces and the research question of the study are relevant research limitations.

The information available depends partly on whether a visitor is anonymous or identified. Session-based signals can be obtained without knowing the customer's identity, whereas purchase history, previous visits or even customer-level attributes usually depend on some form of recognition. Hendriksen, Kuiper, Nauts, Schelter and de Rijke (2020) distinguish between features available for anonymous sessions and additional historical features used once customers have been successfully identified, based on more than 95 million sessions from a European e-commerce platform. The implications of these two approaches are practical as well as ethical. While history helps to construct a rich profile of a customer, it may also restrict model coverage and pose a risk to the customer's privacy. A model for anonymous visits will therefore have to rely only on information it can draw from what it observes during a specific visit.

At an even greater scale, Wen, Lin, and Liu (2023) analysed approximately 33.3 million clicks obtained from 445,336 anonymous sessions and also used behavioural trends and product popularity to model purchase intention. In a similar approach Requena et al. (2020) also predicted purchase outcome from a simplified clickstream trajectory, i.e. with limited information about browsing. These indicate the opportunity to predict, with some success even when detailed customer history or even product meta data are not provided. Both approaches, however, depend on event definitions, methods of collecting and predicting window specific to their own platforms. The predictive capabilities achieved by one online or land retailer may therefore not extend to others with their different business processes and/or target clientele - as would often be the case in an online environment where each online platform or even products require a unique interaction set.

Marr (2016) uses the Amazon case to provide a professional example of the use of behavioural and transactional data in forming a picture of the customer. He gives examples of the use of purchases, browsing activity and reviews, contextual information and patterns among similar customers for product recommendations and for developing a broader view of customers (pp. 289–290). These observations remain useful in explaining how multiple behavioural signals might support personalisation, although the case presented is a historical account rather than technical documentation or causal evidence of business performance. It also highlights questions about data integration, transparency and the limits of customer profiling.

Clickstream records are useful only when the context in which they are being analysed is known. Session boundaries should be consistent, and each variable must be available at the time of prediction. Furthermore, observed behaviour should be interpreted in the commercial environment. Just seeing a pattern does not mean that the firm should respond to it. Purchase prediction moves the analysis one step further by using these behavioural signals to predict whether a particular session will result in an actual purchase.

3.3 Predictive Analytics and Purchase Intention

Predictive analytics finds patterns in historical data to estimate an outcome not known yet. In the present case, the task is labelled as predicting purchases from historical data. More specifically, previous sessions are labelled as purchases or non-purchases, making this a supervised binary classification problem. From this information, a class or probability can be produced for a new session. However, there is a clear distinction that should be applied here, although the literature commonly refers to this task as purchase-intention prediction. When the target records whether revenue was generated, the prediction essentially concerns the customer's observed decision and not the intention per se. This distinction was pointed out by Cirqueira et al. (2020) in their work as well. They separate the targets into three categories, namely customer intent, buying-session and purchase-decision prediction, since they do not represent exactly the same problem.

The modelling process starts before an algorithm is selected, with the definition and labelling of sessions and their transformation into features which can be used consistently during training and prediction. What is included in these features will depend on the data available. A session might, for example, be described by the pages opened and how long the visitor remained on them. Information about where the visit came from and the type of visitor can be included as well. Data are then divided such that the model is trained on one subset and evaluated on observations which were not used to build the model. Testing with previously unseen observations gives some idea of what might happen with future sessions. There remains only one test, however. Results may change for another retailer, in another market or after customer behaviour changes over time.

A point in design is at which moment the prediction should occur. A model evaluated after the complete session has access to richer evidence, including behaviour right before the transaction. It may be useful for some kind of retrospective analyses but is too late to support intervention. On the other hand an early prediction model works on fewer events while the customer is still browsing. Using simplified clickstream trajectories, Requena et al. (2020) were still able to predict purchases from short observation windows. Hendriksen et al (2020) also show that depending on whether the visitor remains anonymous or becomes identified the available predictors will change. Not only the amount of information but whether each variable could legitimately be used in an operational setting is determined by the prediction point.

Decision trees are a possible form of classification. Trees divide up observations using a sequence of rules based on feature values, giving rise to an intuitive structure understandable and explicable to non-technicians. This interpretability may be relevant in explaining to a retailer why certain sessions received different predictions. Trees capture non-linear relationships without restricting each predictor to have a uniform effect. The fact that trees are simple to read is not to say that they can be trusted, however. Unrestricted trees may pick up noise in training data, resulting in unstable rules that perform poorly on new observations. Depth, conditions on minimum samples required, and use of pruning affect complexity of the model and generalisation ability.

There are several attempts to solve modelling problems based on the Online Shoppers Purchasing Intention dataset utilising various techniques in previous research. Baati and Mohsil (2020) used random forest modelling together with methods for handling class imbalance using this dataset. Necula (2023) worked on this dataset in a wider study of online consumer behaviour. The above studies should not dictate the design of the practical application. Their headline performance figures are difficult to compare directly due to the possible influences of data pre-processing, data splitting techniques, resampling methods and specific definitions of performance metrics. Only one decision tree is considered here because it demonstrates the taught method from data preparation to interpretation, while helping to keep the exercise simple and interpretable.

The prediction model created for this assignment therefore needs to do more than satisfy its statistical objective. The model cannot use information that will only become available after the decision. Its rules should also make sense to the people who will use the result. The outputs it produces must also correspond to actions that can realistically be taken by the decision-making organisation. But classification performance does not show that a specific intervention will raise sales, so this would have to be tested directly, in something like a controlled business experiment. Therefore purchase prediction can perhaps best be regarded as being an aid to decision making instead of proof that a particular commercial activity will succeed.

3.4 Business Importance and Organisational Impact

The ability to predict a purchase by a visitor should also aid e-commerce decision-making by enabling companies to separate those sessions that need more attention from those that will proceed on their own. For a session with stronger purchase signals, the response might be product assistance, a recommendation or an incentive. Another session can simply continue without intervention. The purpose of this prioritisation is a more careful utilisation of resources for promotion, as across-the-board personalisation may result in lower margins or disturb customers who did not need assistance. Akter and Wamba (2016) discuss several uses of e-commerce analytics. Personalised offers and dynamic pricing are among them, together with customer service and support for business decisions. These examples indicate where value may be created. A prediction alone is not enough reason to respond automatically.

The possibility of understanding consumers better through clickstream analysis extends to browsing patterns; a completed purchase is only partially indicative of user behaviour. Sales show what has been bought, but with more extensive browsing information one also sees abandoned journeys, frequent comparisons and visits involving products which do not get purchased. Insights derived from this information may be applied in website design, recommendations or campaign-targeting. For example, Sakalauskas and Kriksciuniene (2024) employed clickstream engagement to identify high-value users and examined its relationship with campaign click-throughs and conversions on two e-commerce websites. Therefore, clickstream data may be used to identify areas for improved targeting, although identifying high-value customers is not necessarily the same as predicting whether an individual session will bring in revenue.

Marr’s case on Amazon demonstrates the use of behavioural and transactional signals to build profiles and recommend products (Marr, 2016, pp. 287, 289–290). In the eyes of the retailer, the value of recommending goods is that the effort customers make when searching for goods is reduced and the relevance of products they might be interested in is increased. However, in Marr's case, there is no isolation of the effect that recommendation technology has on Amazon's growth; nor is there a financial return specified from the use of such technology.

Whether an insight translates into impact on an organisation depends on what happens after it is produced. Predictions need to reach the right process, be understood by decision-makers and lead to an action that can be successfully implemented at the right time. Research evidence from Wamba et al. (2017) suggests that Big Data Analytics capability influences performance in part through process-oriented dynamic capabilities. It provides further support to the point that data and models do not create an advantage independently of people. Skills, system integration, managerial responsibility and the ability to adjust operations define how well an analytical output is utilised. Hence, a smaller retailer may obtain limited value from a technically advanced model if it lacks the processes or resources required to act on its output.

A predicted purchase probability does not mean that an intervention led to a sale taking place. Some customers may already have bought unaided, while others may respond negatively if sent an unnecessary message or incentive. Measures should focus on evaluating actual business value gained through incremental conversion, margin, intervention cost and customer response, preferably against a control group.

3.5 Data and Methodological Challenges

The predictive model generated from any company's data can only be as good as the data entered; clickstream records could contain repeated entries, inconsistencies in session definition, automated entries through web crawlers or events resulting from changes in design rather than from consumer behaviour. Similarly, variables created through aggregation can produce further unreliability as the same value could be masking contrasting journeys. A session that was recorded to have lasted a long amount of time could be indicative of engaging material that led consumers to explore the website willingly, but could just as easily signify difficulty with finding what they needed despite numerous clicks. Liu et al. (2026) identify data quality, together with interpretability and privacy, as issues in consumer Big Data research. Data preparation should be integrated with analytical thinking rather than treated only as a pre-step.

With class imbalance, purchase prediction faces a special challenge because purchasing sessions only constitute a small minority of all samples. With the majority class dominating, a model can yield good results overall while performing badly on the outcome that the company is more concerned with. Baati and Mohsil (2020) found that the untreated classifiers tended towards the majority class and applied SMOTE to the training set. Resampling could enhance the model's awareness of patterns in the minority, but this shifts the training distribution and may even generate instances that are not based on real customer journeys. Its necessity should not be presumed, and the test set should be left with a genuine class distribution for evaluation.

Care should also be taken when choosing metrics. Accuracy gives the percentage of correct answers while hiding how errors are distributed between purchases and non-purchases. Precision shows how many of the events predicted as purchases were actually purchases, and recall shows how many of the actual purchases were detected. F1 score combines the two whereas it is sometimes the case that businesses find one (precision or recall) worse to miss than the other. Balanced accuracy calculates the average recall of the two classes, giving each class the same weight; for this reason, it is less sensitive to the larger number of negative cases than overall accuracy. Brabec, Komárek, Franc and Machlica

(2020) show that precision tends to move with the ratio of positive to negative samples; the same can be said of F1-score and the area under the precision-recall curve; classifiers may even reverse their original ordering when prevalence changes. Therefore results should not be summarized as one headline, but be given with prevalence, a confusion matrix and together with other complementary measure results. The same applies for all performance results.

Another issue is whether each feature is actually available when it is needed. Variables available towards the end of a session can be associated with purchase and reflect the behaviour that led to conversion at a late stage. These can improve test results and limit the usefulness of such models for intervention. PageValues in the selected dataset is one such variable, standing for the average value of the pages visited before an e-commerce transaction is completed (Necula, 2023). It may be predictive without directly leaking future information, but the procedure and reference period employed in its calculation need to be known before including this variable in the model. The timing matters here. If the events used to calculate PageValues have not happened yet when the model makes its prediction, the evaluation may make the model appear more useful than it really is. A sensitivity analysis can be conducted without this variable to establish how much the conclusions depend on it.

Outliers, such as unusually long page durations, high visit counts or rare traffic patterns, produce another tension. These may occur because of errors, as discussed before, but they may also represent interesting, though unusual, customer behaviour. Removing them through strict thresholds could filter out commercially relevant sessions. Necula (2023), while working with the same public dataset, opted to retain outliers because they might capture meaningful behavioural variation. This choice can be defended, but improbable values still need to be investigated and any treatment applied should be recorded. The decision to exclude, transform or retain the values should not be left to a fixed statistical rule, but should follow an explicit rationale supported by logic.

Limitations around generalisation of the models come from the context in which the data were collected. A model trained on only one retailer will pick up relationships shaped by its catalogue, website UI, customer base and measurement process. Although Hendriksen et al. (2020) gathered data from more than 95 million sessions, these came from a private platform during a four-week period. Furthermore, Requena et al. (2020) performed their research in a different fashion-related e-commerce setting, with a different way of representing activity events. Their scale and realistic settings strengthen the individual studies. However, neither study guarantees that the patterns identified will apply to another retail company. Public datasets support reproducibility, but their aggregated and historical nature may make them less similar to current real-world operating environments.

Relationships might change after deployment too. Promotional campaigns, seasonal demand, redesigned interfaces and changes in customer behaviour might affect the prevalence of purchases within the dataset, as well as the relevance of predictors. The same model which performed well during development might prove less reliable later, yet none of its technical components might have changed. Monitoring should focus not only on overall accuracy but also on changes in class distribution, error patterns and behaviour of certain features over time. Retraining of the model might be warranted, although frequent updates may complicate explanations and governance.

What a model score means depends on what the model is meant to do. The issues discussed above can change that meaning in different ways. Poor-quality data and class imbalance may distort the result. A feature that arrives too late or an outlier handled badly creates another problem. Customer behaviour may also change after deployment. Finally, even an intelligible decision-tree rule can be based on poorly timed or unsuitable evidence. These decisions do not end with model performance. They can create ethical problems too, and these are considered next.

3.6 Ethical, Privacy and Governance Challenges

Organisations running clickstream analysis can create value from behaviour that their customers may not realise is being recorded or combined. Page views, time spent browsing, traffic sources and sequences of navigation clicks may look harmless when considered alone; together they can assist organisations in making detailed inferences about customer preferences and propensity towards making purchases. The absence of a customer's name does not remove this concern. There are ways of profiling and distinguishing anonymised sessions and potentially linking them to other identifiers. The aspect of privacy should be assessed by reflecting on what the data enable an organisation to deduce and do, as opposed to whether or not a record itself contains directly identifiable information.

Tension is also illustrated using the example of personalisation. Customers may enjoy the benefits of more relevant recommendations that require less search effort, or of getting help at the right moment. But the same practices can make customers uncomfortable as they might feel like they are monitored too closely. Canhoto, Keegan and Ryzhikh (2024) explore the tension of AI-enabled personalisation in physical retail in their qualitative study through the idea of a personalisation-privacy paradox. Customers may appreciate a relevant and timely offer but may want to have more control over their personal and location information being used to form it. This is not simply a choice between personalisation and privacy. Their responses depend on the type of data used, the transparency of the practice, the context of the interaction and whether the message generated appears proportionate.

There are some recent experiments showing that more detailed personalisation is not necessarily more effective. In their 3 × 2 experiment with 360 participants, Kim and Han (2025) compared contextual personalisation with highly intrusive personalisation based on personally identifiable information. They discovered that intrusive personalisation became less effective once the experimental setting was designed to prime privacy concerns, suggesting that moderate personalisation might work better. The findings of Kim and Han provide causal evidence for a possible backfire effect; however, some features of the research should be considered when interpreting the results. Participants in the study were in South Korea, and the experimental design partly combined variations in both the amount and type of personal information involved. Nevertheless, their work calls into question the assumption that collecting more customer data will consistently improve marketing outcomes.

Transparency and consent are major governance issues, though of course, just presenting a privacy notice will not ensure that the user understands it. They may agree, not knowing or imagining how the behavioural traces will be used, combined and retained, or how they may affect their user experience. Data minimisation, on the other hand, is a more practical safeguard - one only collects and retains data to achieve a specific analysis outcome and does not include variables for no other reason than because they can be. Privacy and security, alongside informed consent and governance, are listed by Akter and Wamba (2016) among the main challenges facing e-commerce analytics, linking responsible use of data to sustainable business value.

Another concern with behavioural prediction is unequal treatment. If one of these models determines which customers receive help, recommendations, discounts or attention, differences in access, buying power or previous marketing exposure found in historical behaviour may be repeated through model-based treatment. The chosen dataset lacks conventional demographic variables, reducing some of the risk on this front, but it cannot fully eliminate proxy effects; the model can still divide its users into categories according to traffic source, type of visitor, region or browsing patterns, which may then be treated differently. Again, interpretability can help identify such rules, but a transparent model is by no means inherently fair.

Governance thus needs to cover the entire analytical lifecycle. Data ownership and boundaries of access should be established, security controls should be tailored to the varying sensitivity of behavioural profiles, and the purpose of the model should be documented prior to deployment. Companies also need to decide who can authorise changes, investigate surprising outcomes and suspend the model when necessary. Cirqueira et al. (2020) point to interpretability, privacy regulation and evaluation across multiple platforms as continuing research issues in e-commerce purchase prediction. These aspects show that governance cannot simply be inserted after a model is developed; it affects the collection of data, the process of evaluating model performance and the actions that are authorised.

Marr (2016) uses Amazon to show why integrated customer profiles can be commercially useful and, at the same time, sensitive. In the case described, purchases, browsing behaviour, reviews and location data were brought together with information from external sources to build a broader picture of each customer (pp. 289–290). This wider view could support better recommendations, but it could also reveal more about a person than they knowingly shared in a particular transaction. The example is historical. It reflects practices reported in 2016 and should therefore be read as an industry illustration, not as a description of Amazon's current systems, services or governance.

Anonymising the data and keeping it secure are only part of responsible purchase prediction. The organisation must also have a legitimate reason for using the model, collect no more data than the task requires, explain how the resulting intervention works and continue checking its effects once it is in use. Higher conversion does not necessarily mean that the model creates lasting value. If customers are profiled without sufficient justification or begin to lose trust, an immediate commercial gain may later become a cost to the wider organisation. Ethical judgement and effective governance are therefore not separate from business value. They help determine whether that value can be sustained.

3.7 From Analytical Results to Business Value

Decisions, not data, give birth to analytical value. Before creating a purchase-prediction model, the organisation needs to find out what it is for, when its output is needed and who can act on the insights. A model that supports a customer during a session would use different data and infrastructure from one based on past journeys or used to create remarketing audiences. Unless it is tied to a decision, such a prediction model might only achieve success at a technical level.

Marr (2016) argues that the ideal process is to proceed from business strategy to the questions that need to be answered to improve organisational performance. Relevant data should then be acquired and analysed because they help answer the questions set, rather than simply collecting all available data with the vague expectation that they will eventually result in insight (p. 294). This principle also applies to clickstream analytics, where large numbers of events can be collected at a relatively low cost. However, collecting every available interaction may increase storage and governance demands without improving the decision. The focus should be on data which are timely, reliable and proportionate to the intended use.

Beyond this, operationalisation might entail integration into existing processes and systems when predictions are to be ingested by a website, fed to a recommendation service, communicated by a customer-support agent or pushed to a campaign platform within a required time window. It should also be clear who holds responsibility for the action which the model generates, particularly when prices, incentives or the treatment of individual customers are involved. According to Necula (2023), conditions which apply to the use of machine-learning applications in e-commerce include data accuracy, technical know-how within the company, computational resources, integration with existing systems and the measurement of return on investment. Clearly, deployment amounts to an organisational change, as it is far from being only the last step of coding.

For this reason, business evaluation should be distinguished from model evaluation. While measures such as precision and recall indicate whether the classifier distinguishes those who purchase from those who do not, they do not indicate whether action based on the model's output improves business performance. An intervention may increase conversion but decrease margin, or provide discounts to customers who would have purchased anyway. Relevant business measures depend on the use case; these may include incremental conversion, intervention cost, profit contribution, customer response and longer-term retention. Controlled experiments could help analyse the impact of the action taken against the likelihood that the customer would have made the purchase anyway.

For the link between analytics capability and firm performance, it also matters whether and to what extent processes can be adjusted by the organisation. Wamba et al. (2017) determined that process-oriented dynamic capabilities partially mediate this relationship, suggesting that insights create value by fostering an organisational response. Given their cross-sectional methodology, however, this does not imply that such a return applies to all organisations. Firms differ in their data maturity, capability set, scale and ability to implement change. As such, the same model may lead to different firm-level outcomes.

The final step is to monitor the value delivered after deployment. Purchase prevalence, browsing behaviour and campaign conditions might change, thereby affecting the model's performance and the economics of an intervention. Monitoring needs to cover technical shortcomings, business performance, customer reactions and unintended effects, rather than assuming that a previously successful model will continue to serve its purpose. Viewed more broadly, business value depends on the connection between strategy, suitable data, modelling, operational action and governance. This connection provides the basis for the following discussion of the main tensions found in the literature and for the proposed decision-oriented approach to purchase prediction.

4. Discussion and Suggestions

The paper argues for purchase prediction to be viewed as a decision-support process and not as a purely technical classification problem. Some frequent issues mentioned in the literature emerge, though their importance will depend on the business decision, at what point the prediction is needed and the consequences of acting on its output. This section discusses a few of these issues from the point of view gained from the literature review and proposes a practical approach for balancing predictive performance, business value and the responsible use of data.

4.1 Critical Synthesis

There are five competing and related tensions in purchase prediction that companies must consider: early prediction and richer information, predictive performance and interpretability, personalisation and privacy concerns, technical performance and business value, and data scale and relevance. These need to be balanced in relation to the business decision being made.

The first tension is about early prediction and richness of information. Additional clicks and behavioural signals later in a visitor's session may improve classification performance, but the longer the system waits, the less time there is to affect the final outcome. Requena et al. (2020) illustrate this trade-off between prediction time and information richness by comparing conversion predictions based on clickstream sequences of different lengths. Extending the sequence improved performance further, although short sequences already produced useful predictions. The appropriate point of prediction depends not only on performance but also on the intervention being considered. A recommendation may remain useful relatively late in the session, but proactive assistance or a tool for averting abandonment might have to operate earlier, in which case a less certain prediction may still be useful.

Another source of tension comes from trading predictive power with interpretability. While more complex models may be able to discern patterns that simpler approaches miss in large and sequential datasets, it may still be difficult to explain, challenge or translate their output into intelligible sets of operational principles. Requena et al. (2020) reported some benefits from more advanced architectures, although they pointed out that these came with greater computational and infrastructural demands. Whilst a decision tree could probably not promise the best possible performance, its operating rules can be investigated by managers and analysts. Moreover, interpretability does not have equal value in all settings, but it becomes important where predictions influence the treatment of customers and justification of an action might be needed. The underlying issue is not whether simpler models are always better, but whether additional complexity leads to an improvement sufficient to offset the operational cost and the loss of transparency.

Personalisation creates the third tension. Behavioral data can make recommendations and interventions more relevant, but at the same time, increased detail may make customers feel monitored. Evidence from physical retail shows that AI-enabled personalisation can create perceived benefits while also raising concerns related to privacy (Canhoto et al., 2024). Further to that, the backfire effect shown by Kim and Han (2025) indicates that intrusive personalisation can become less effective when privacy concerns become salient. Therefore, in practice, maximum personalisation should not be taken as a goal. Sometimes, a restrained intervention based simply on the session context can produce more value than a specific message built from extensive customer history.

A fourth tension exists between technical performance and business value. Accuracy, recall and related measures describe the behaviour of the classifier rather than its commercial impact. Knowing that a prediction is correct does not show whether the intervention was needed or brought business value. Customers classified as likely to purchase something may do so anyway; the discount is then wasteful. Conversely, a false positive may lead to an intervention that still improves the customer's experience. Model evaluation should stay connected to, but distinct from, business evaluation. Predictive measures verify whether the model identifies patterns; controlled business measures then assess what happens when the organisation acts on those patterns.

The last source of tension is the scale and relevance of data. Large numbers of events in a clickstream may give a rich account of activity, but they might also be redundant, too old or, more importantly, poorly defined. Smaller datasets with proper governance can be more helpful, especially if they are carefully specified and their variables help explain the business decision at hand. This issue is relevant to the practical study. The public dataset is useful for transparency of analysis, but it does not reproduce the scale, speed or technical architecture of a large e-commerce platform. It is useful for illustrating a method and bringing some analytical choices to the surface, but it cannot be presented as a Big Data deployment.

All these tensions reinforce the idea that purchase prediction is a socio-technical decision process. Optimising components in isolation can weaken the overall result: later data may improve accuracy but remove the opportunity to act; greater personalisation may increase relevance but reduce trust; and a more complex model may perform better while becoming harder to understand and govern. For this reason, the preferred approach is not to maximise data collection or model complexity, but to select the level of information, performance and intervention that is proportionate to the decision being supported. The following framework translates this position into a more structured set of stages.

4.2 Proposed Decision-Oriented Framework

The proposed framework is based on the issues pointed to during the review. A six-stage decision-oriented framework for the use of purchase prediction in e-commerce is proposed. This is designed to reduce the risk of modelling decisions taken by online businesses getting out of sync with business needs, operational timing and careful data use. The framework has been developed for this paper and has not been empirically validated; it represents a structured synthesis of the reviewed literature. As such, it should be treated as a workable decision-oriented framework that would require further testing in organisational settings.

1. Define the business decision. The project should begin by determining the decision the prediction will inform. Predicting whether a particular session will lead to revenue is not, by itself, a complete business goal. The business should ask: does it want to provide assistance, recommendations or incentives during a session? Does it want to remarket sessions that did not generate revenue? What kind of decision needs to be made based on the prediction output? It should also be understood who within the business will use the prediction and what outcome is expected from it. This is to avoid building up a clever, sophisticated prediction model which does not have any practical application.

2. Establish the prediction point. The organisation has to decide when the prediction must be made. That decides what behavioural information can legitimately be used. While an earlier prediction gives the organisation more time to act, it has only limited evidence at its disposal. A later prediction, on the other hand, is richer in evidence but does not allow much time for action. No variable recorded after that decision point should be employed in the prediction, even if it drastically improves performance on the test set. Feature availability must be according to the operational cycle and not according to convenience while the model is built.

3. Select available and legitimate data. Only data that are relevant, reliable and proportionate to the stated purpose should be included. At this stage, the quality of data, definitions of a session, missing values, unusual observations and the meaning of aggregated variables should be considered. It also involves ethical judgement: there are variables that might be technically available but unnecessary or intrusive for the specific decision. Data minimisation reduces exposure to privacy risks, while a model built with only the data needed can be easier to understand and govern.

4. Evaluate technical and business performance. When assessing potential techniques on a technical level, it is wise to address class imbalance and the consequences of different errors. Accuracy alone does not provide much information; a look at the confusion matrix, precision, recall, F1-score and other balanced measures provides a view into what the classifier is actually doing. These results then need to be analysed in terms of the business case. False positives may be costly in terms of unnecessary interventions, while false negatives may represent missed opportunities. The most desired model will not necessarily be the one with the highest metric, but the one whose error profile is acceptable for the planned action.

5. Operationalise with oversight. The generated prediction needs to be fed into a process which knows how to use it within the required time. There need to be people responsible for approving the model, interpreting its output and reacting to unexpected effects. Automatic actions could be adequate if the risk is low, whereas pricing, customer eligibility or sensitive forms of targeting should have stronger human oversight. The model's purpose, allowed data, shortcomings and conditions for suspending its use should be documented.

6. Monitor value, change and unintended consequences. Following implementation, monitoring continues. As customer behaviour, purchase prevalence or website design changes, the technical performance of the model may degrade. Business outcomes should also be monitored to examine whether an intervention actually creates incremental value or merely accompanies purchases that would have happened anyway. Along with conversion and cost, customer reactions, privacy concerns and unequal treatment should also be monitored. When sufficient evidence no longer supports using the model, it should be revised, retrained or withdrawn.

The proposed framework identifies six separate stages; they are, however, not a one-way sequence. Monitoring might show that the business decision, prediction point or data allowed for use needs to be adjusted, in which case the process returns to an earlier stage. In this way, purchase prediction is considered an ongoing organisational capability rather than a once-and-for-all model which remains intact forever. Connecting choices about techniques with decisions, actions and governance responsibilities gives the framework a practical basis for using clickstream analytics.

4.3 Practical Recommendations

Instead of going the whole way to creating a customer-prediction system at once, e-commerce organisations should take a step back and start modelling with a narrow and measurable use case. For example, the use case may be to identify sessions eligible for an offer of optional live assistance (e.g. to improve conversion rates) or to select customers for part of a seasonal remarketing campaign (e.g. to stimulate purchases). The business decision, target groups and expected outcomes must be defined as modelling inputs. This makes judging whether the prediction actually offers a solution to a real problem easier and limits the risk carried by the e-commerce business when rolling out an untested system.

Intervention policy should be kept separate from the model. A predicted purchase probability does not dictate that a particular discount, recommendation or message is appropriate - the costs and consequences of various actions differ, so their respective thresholds may also differ. For example, a low-cost suggestion of a complementary product could be offered more widely to customers than a financial incentive; a greater onus could (and should) be placed on the financial incentive to show the incremental value it creates. Thresholds should reflect these economics and customer consequences rather than solely being tweaked to improve a technical metric.

Feature timing should be tested explicitly. A variable may make the prediction stronger and still arrive too late in the session for much to be done with it. The decision tree on the selected public dataset should be tested with the full set of planned features and then with PageValues excluded as a sensitivity analysis. Both tests still use the same analytical method. Although, as discussed above, PageValues may not be leakage in this dataset, it does provide insight into how dependent the results are upon a strong and potentially late variable. In the end, calling it leakage comes down to a practical question: could PageValues have been calculated using information already available when the prediction had to be made?

The test distribution should reflect the naturally occurring class distribution. Resampling can occur when training the model; however, the sample to be evaluated should not be resampled and must represent what the organisation expects to encounter in relation to purchase prevalence. Results must include a confusion matrix, precision, recall and the F1-score, as well as balanced accuracy alongside overall accuracy. False-positive and false-negative classifications should, more importantly, be discussed in relation to the consequences of the proposed intervention. This makes the evaluation understandable to managers instead of portraying one metric as an all-encompassing measure.

Application design should aim to embed the responsible use of data within the applications themselves; i.e. restricting the use of behavioural variables to the stated purpose, controlling access, retaining them when justified and not presuming that anonymity implies safe use. This is where anonymity can become a risk: when combined with device or location or perhaps combined with other behavioural or historical data. All customer-facing messages or interventions must remain proportionate to the session context. Highly tailored messages derived from inferences, which may appear unexpected, may ultimately serve to erode trust, regardless of the statistical accuracy of the prediction.

The system should be deployed gradually and compared against a clear baseline. With a controlled experiment, it is possible to estimate whether applying a model-driven intervention improves outcomes compared with current practice or doing nothing. Conversion alone may not be adequate, however; margin, intervention cost, opt-out behaviour and customer response can also be estimated to see whether other effects are hidden by sales (a short-term measure). As the model may also affect who receives offers or assistance, the distribution of these actions should also be examined for potentially unjustified differences between the various customer groups.

Finally, monitoring should have defined owners and stopping conditions. Changes in website design, campaigns or customer behaviour may make previously useful rules unreliable. A model should be reviewed when class prevalence shifts, errors increase or the business value of interventions declines. Privacy complaints, unexpected targeting patterns or evidence of customer harm should also trigger investigation. These recommendations favour a limited, explainable and measurable application over immediate automation at scale. Such an approach is more likely to produce credible evidence of value while preserving the organisation's ability to correct mistakes.

5. Conclusion

This paper analysed how clickstream data and predictive analysis methods can be harnessed for online purchase predictions, considering the commercial, technical and ethical aspects influencing the usefulness of such models. Behavioural data records can indeed contain signals of how a session may unfold, such as navigation patterns, page engagement, context or page-specific characteristics, amongst others. Once these data are identified and cleaned, classification methods can help to distinguish the sessions more likely to generate revenue. However, their usefulness largely depends on whether suitable opportunities to act on that prediction are still available to the organisation.

Purchase prediction needs to be treated as a decision-support process, not only as a technical problem. Technical performance metrics show whether a classifier distinguishes actual conversions, while incrementality, margin, cost and customer response determine whether acting on its predictions creates value. There are also business factors affecting the value of purchase predictions. When a predicted probability is followed by a conversion, this does not show that acting upon the prediction caused the conversion, and thus its impact needs to be computed against a credible baseline.

Several methodological considerations limit the conclusions that can be drawn from clickstream models such as the simple decision tree used in this study. Weak performance for the relatively small number of purchasing sessions may become hidden by class imbalance, whereas variables that become available late or are not fully understood may give optimistically good results even when their practical utility is low. The treatment of outliers, the session definitions, or indeed when and how the data were prepared will all influence the patterns learned by the selected model. Performance achieved on the historical dataset used in the case study cannot be assumed for retailers of different sizes, other markets, or indeed different time periods. A readily available public dataset was used to illustrate the concepts through an example decision tree due to its transparency; however, it does not represent the scale, velocity or infrastructure of a large e-commerce platform.

The analysis did not show that more data or personalisation is automatically better. The value of being given accurate insights into behaviour can sometimes come at the cost of personal space or control over how a service provider uses the data, especially if those insights result in customers not being treated fairly. The data might be anonymised, but it would not be unreasonable for behavioural profiles to be derived from those records and thereby provide an even "more personalised" experience. Therefore, when data are not handled properly, customers may feel uncomfortable and vulnerable even if the practice is presented in an understandable manner. Responsible data use requires a legitimate purpose, a proportionate amount of data gathered and some form of governance and monitoring of how the customer is affected. They do not lie outside value creation from data analytics; instead, they determine whether an analytical application will remain trustworthy and sustainable.

In addressing this combination of concerns, the paper introduced a six-stage decision-oriented framework, defining the business decision, establishing the prediction point, selecting available and legitimate data, evaluating technical and business performance, operationalising it with oversight, and monitoring value, change and unintended consequences. The framework is the main contribution of the paper. It is not empirically validated, but forms a bridge between the modelling choices and the organisation's responsibility for the decision.

In answer to the research question, clickstream data and predictive analytics can support purchase prediction by converting observable digital behaviour into timely estimates that inform selective business action. Effective use is conditional rather than automatic. It requires relevant data, appropriate evaluation, realistic timing, organisational capability and responsible governance. The main lesson is therefore not that firms should maximise data collection or predictive complexity. They should develop the simplest credible application that supports a defined decision, produces measurable incremental value and remains proportionate to the customer relationship.

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